Summer Training Report

On

MICRO-ELECTRO-MECHANICAL SYSTEMS(MEMS): COMB TYPE MICRO-ACCELEROMETERS

At

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Under the Guidance of

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1.INTRODUCTION

MEMS accelerometers are used to sense the acceleration experienced by a system. They have been good examples for MEMS commercial products and have made their way into most of our daily lives. The most common uses for MEMS accelerometers are in airbag deployment systems for modern automobiles. In this case accelerometers are used to sense negative acceleration of the vehicle. A processor will analyze the magnitude of the acceleration and decide on whether or not to deploy the airbags in the vehicle. MEMS accelerometers are quickly replacing conventional accelerometers for airbag deployment systems in automobiles. The reason behind this increasing popularity is, the MEMS accelerometers are much smaller, lighter, more reliable and are produced for a fraction of the cost of the conventional bulky accelerometers. Several new innovations in micromachining have been combined to make a commercially available accelerometer for low g (gravity acceleration) applications. One of the successful innovations is straight beam comb accelerometer. The straight beam is anchored at four points. While this sensor is stiff and robust, it is also sensitive to mechanical stress imparted to the die from the package and die amount. The recent innovation of Analog devices Inc. ADXL50 is folded beam comb accelerometer. The folded beam structure is anchored only at two points. The wrap around structure relieves the tension of poly-silicon which makes the beam much less sensitive to package stress. At the same time, the compliance of beam, i.e. the deflection of beam for a given acceleration applied, is increased which enhances the sensitivity. In addition to accelerometers, many other MEMS devices such as MEMS gyroscopes, digital micromirrors, MEMS optical switches, DNA chips, have been successfully
developed and commercialized. MEMS, together with nanotechnology, are believed to be the drive to trigger the next wave of technology revolution.

As MEMS technology continue to grow, MEMS device design optimization is becoming an interesting and important research issue. In order to design a MEMS device to meet the given specifications, the relationship between the device performance and various design parameters must be investigated. Various efforts on MEMS device design optimization and automation have been made. For example, in the design optimization and simulation on a micro-electromagnetic pump was discussed. In the mechanical design and optimization of a capacitive micromachined switch was proposed. In an automated approach is used to generating novel MEMS accelerometer configurations. Considering the commercial success of MEMS accelerometers, the design optimization of a folded-beam MEMS comb accelerometer device is discussed in this paper. The relationship between the device sensitivity and the design parameters (such as beam width, beam length, mass width) is analyzed. ANSYS FEM simulation is used to derive the device sensitivity for various design options.

2. Device Design

The structure design of a poly-silicon surface-micromachined MEMS comb accelerometer is shown in Figure 1. This device is similar as ADXL 150 accelerometer developed by Analog Devices Inc.

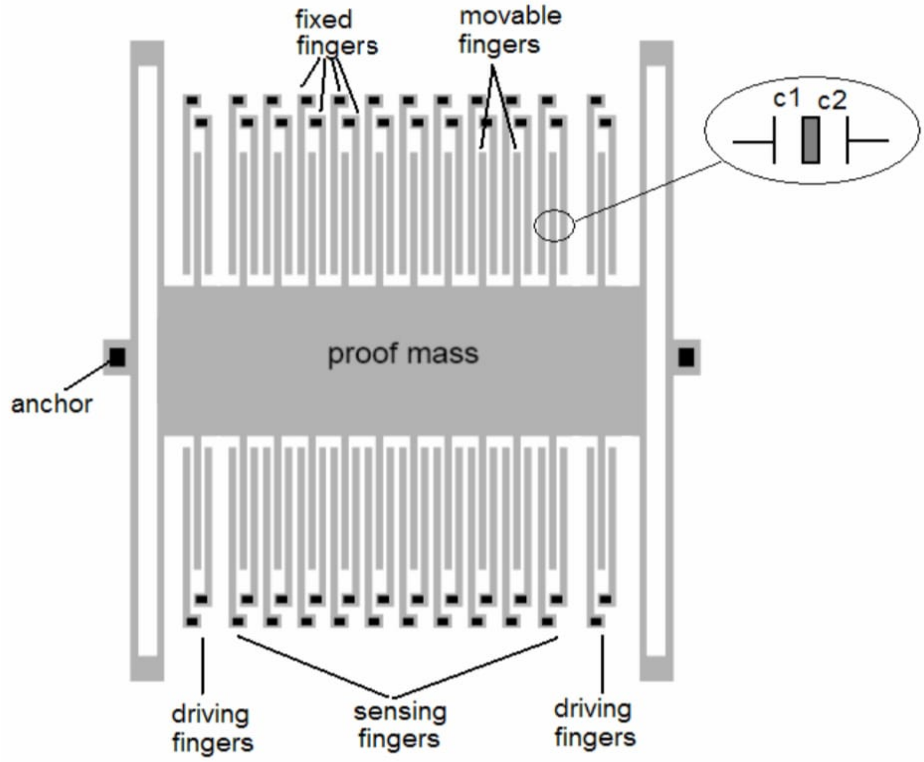

Figure 1. Structure diagram of folded-beam accelerometer

The movable parts of this MEMS comb accelerometer consist of four folded-beams, a proof mass and some movable fingers. The fixed parts include two anchors and some left/right fixed fingers. The central movable mass is connected to both anchors through four folded beams. There are many movable fingers extruding from the both sides of the central mass. In the right and left side of each movable finger, there are left and right fixed fingers. The movable fingers constitute the differential capacitance pair C1 and C2 with left and right comb fingers. When there is no acceleration, the movable fingers are resting in the middle of the left and right fixed fingers. In this way, the left and right capacitance pairs C1 and C2 are equal. If there is any acceleration a along horizontal direction parallel to the device plan, the proof mass (M_s) experiences an inertial force – ($M_s \cdot a$) along the opposite direction. As a result, the beams deflect and the movable mass and movable fingers move for a certain displacement x along the direction of the inertial force. The left and right capacitance gaps are changed, hence the differential capacitances C1 and C2 will also be changed. By measuring this small differential capacitance change, we know the value and the direction of the experienced acceleration. This is the working principle of the MEMS comb accelerometer. The comb accelerometer design also supports in-field built-in self-test feature. Among these capacitance pairs, most capacitance groups act as the sensing capacitance and other few capacitance groups act as built-in self-test capacitance. The built-in self-test feature allows the device to be self-tested during in-field usage using electrostatic force. In test mode, when there is no acceleration, a driving voltage V_d is applied to the left or right fixed driving fingers. The electrostatic force will attract the movable fingers toward the left or right direction. By measuring this displacement and comparing with good device response, one knows whether the device is good or faulty. This self-test feature is especially important for the safety critical applications such as automobile airbag deployment.

3. Sensitivity Analysis of the Device

When an acceleration *a* along the vertical direction parallel to the device plan is applied to the accelerometer, the beam deflects under the effect of inertial force. The deflection of beam is in opposite direction of the applied acceleration. The displacement sensitivity of the device is defined as the displacement of the movable mass (and movable fingers) per unit gravity acceleration g (1g=9.8m/s2) along devices sensitive direction. The beam-mass structure of the accelerometer can be treated as a simplified spring-mass model. The four folded-beam can be treated as four springs connected in parallel. For each folded-beam, both sections of the beam can be treated as two springs connected in series. Each beam section can be treated as double-clamped beam model.

Assume for each section of the folded-beam, the beam width and length are W_b and L_b separately. The width and length of central proof mass are W_m and L_m separately. The device thickness (thickness of the poly-silicon layer) is t. There are totally N_f finger groups, among which there are N_s sensing finger groups and Nd driving finger groups ($N_f = N_s + N_d$). For each movable finger, the finger width and length are W_f and L_f separately. When there is no acceleration, the capacitance gap between each movable finger and its left (right) fixed fingers is d_0. The density ρ and Young's modulus E of poly-silicon material are given as below.

The density of poly-Si is $\rho = 2.33*10^3$ kg/m^3
Young's modulus of poly-Si is $E = 1.70\times10^{11}$ Pa
When there is no acceleration, the static sensing capacitance of the MEMS comb accelerometer is

$$C_0 = \varepsilon . N_s . L_f . t \,/\, d_0$$

Where ε is the dielectric constant of air.

Assume there is acceleration along upward direction vertically, the movable mass experiences an inertial force toward right by x, as shown in Figure 2. Assume small deflection approximation (x<<d0), the left (right) capacitances C1 (C2) are changed to

$$C_1 = \varepsilon . N_s . L_f . t \,/\, (d_0 - x)$$

$$C_2 = \varepsilon . N_s . L_f . t \,/\, (d_0 + x)$$

The Differential capacitance change ΔC is

$$\Delta C = C_1 - C_2 = 2\, \varepsilon . N_s . L_f . t \,/\, d_0$$

From above equations we can see that for small deflection approximation, differential capacitance change is directly proportional to the displacement x of the movable fingers.

Further, for small deflection (beam deflection angle < 5˚) , we can treat the accelerometer as simplified spring-mass model. Assume the total sensing mass of the accelerometer as M_s, the inertial force $F_{inertial}$ experience by the sensing mass for acceleration a along sensitive direction is

$$F_{inertial} = -M_s \cdot a$$

Assume the total spring constant of the beams as K_{total}, the displacement x of the movable mass can be calculated as

$$x = F_{inertial}/K_{total} = - M_s \cdot a / K_{total}$$

We can see that the differential capacitance change ΔC of the accelerometer is directly proportional to the experienced acceleration a. Thus by measuring the differential capacitance change, the acceleration can be measured.

The resonant frequency f_0 of the spring-mass system is given by

$$f_0 = ½\pi \sqrt{K_{total}}/\sqrt{M_s}$$

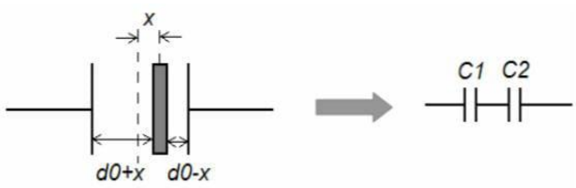

Figure 2. Differential capacitance of MEMS comb accelerometer

According to the literature, several researchers have determined the stiffness constant of the folded beam. Based on CHAE *et al.* [4], and CHAE *et al.* [5], and BOROVIC *et al.* [3], the stiffness constant of a folded beam can be given by

$$k \approx \frac{24EI}{l_{\text{beam}}^3} = \frac{24E}{l_{\text{beam}}^3} \times \left(\frac{t_{\text{beam}} \cdot w_{\text{beam}}^3}{12}\right), \tag{3.15}$$

where E is the Young's modulus, I is the moment of inertia, and l_{beam} and t_{beam} are the length and thickness of suspension beam, respectively.

Based on the above sensitivity analysis we can draw the following conclusions.

(1) We can conclude that the sensitivity of the folded-beam comb accelerometer is inversely proportional to the third power of the beam width W_b, that is, $S_d \propto (1/W_b^3)$ The device sensitivity changes rapidly with the beam width. In other words, beam width of the folded beam accelerometer is a highly sensitive parameter to adjust the sensitivity of the device. A folded beam accelerometer of desired sensitivity can be designed by adjusting beam width. Theoretically we can narrow down the beam width Wb to achieve

very high device sensitivity. However, there is always a bottom limit for the beam width set by the minimum line width in a fabrication process. If the beam width is too narrow (e.g. less than 2μm), it will become very challenging to fabricate the beam because the beam is extremely fragile and can be easily broken. This will greatly reduce the fabrication yield. Thus there is a trade-off between them and we cannot shrink the beam width unlimitedly. In order for high sensitivity, we may rely on adjusting the other design parameters (such as beam length, mass width) as well.

(2) Sensitivity of the folded beam accelerometer Sd is directly proportional to the third power of beam length. That is, $S_d \propto L_b^3$. However, increasing the beam length will also increase the overall device area.

(3). Sensitivity of the folded beam accelerometer S_d is directly proportional to the mass width and mass length. That is, $S_d \propto W_m$ ***and*** $S_d \propto L_m$.

Table 1. Sensitivity Analysis for a MEMS comb accelerometer

Design Parameters	***Dimensions/Performance***
Beam Width, W_b	2μm
Beam length, L_b	290μm
Mass Width, W_m	500μm
Mass Length, L_m	800μm
Length of Movable Fingers, L_f	300μm
Total Number of Driving Fingers, N_d	8
Total Number of Sensing Fingers, N_s	24
Device Thickness, t	30μm
Capacitance gap, d_0	2μm
Material	Silicon
Range	+-10g
Spring Constant, K_{total}	2788.6N/m
Density of Silicon	2330kg/m^3
Young Modulus of Silicon, E	150*10^9 N/m^2

4. Specification:

a. Range

b. Sensitivity

c. Mechanical Bandwidth

d. Non-Linearity

e. Noise Level

A. Range

The range of a comb-type micro accelerometer refers to the maximum span of acceleration values it can accurately measure without distortion or non-linearity. It is typically expressed in ±g, where g is the gravitational acceleration (9.81 m/s²). For example, a range of ±10 g means the device can measure accelerations from –10 g to +10 g. The range is determined by factors such as the mechanical stiffness of the supporting beams, the maximum permissible displacement of the proof mass before pull-in or structural damage, and the linearity limits of the capacitance–displacement relationship.

We are using the RANGE = [-10g, +10g]

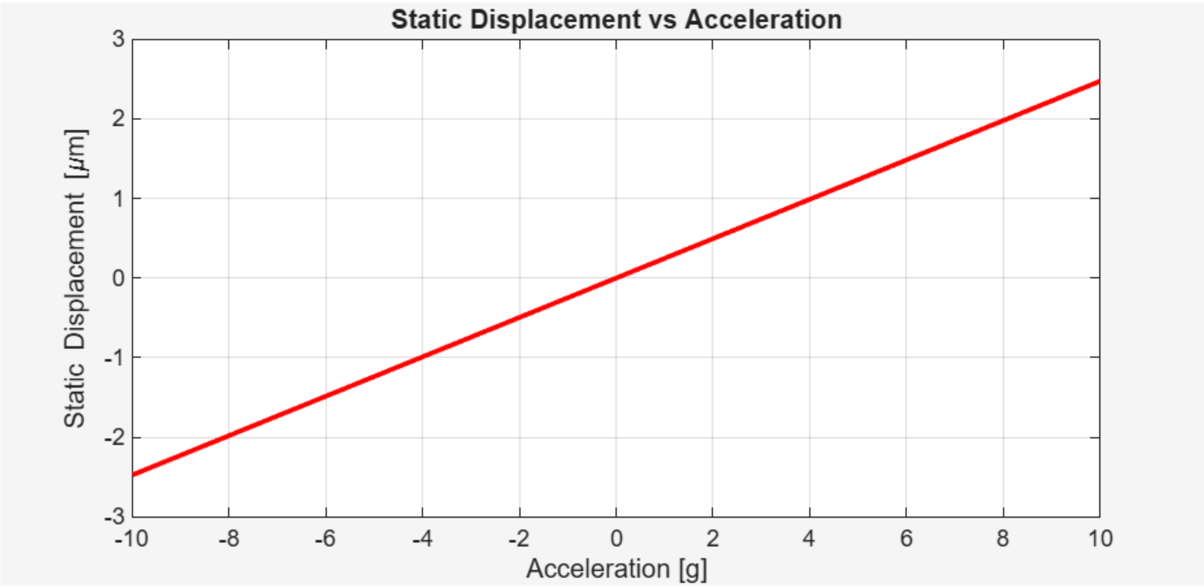

Figure 3: Static displacement vs Acceleration

B. Sensitivity

When the proof mass moves due to acceleration, the gap between the plates changes, resulting in a capacitance variation:

$$\Delta C = C1 - C2$$

The ratio of the change in output capacitance (ΔC) to the applied acceleration (a) is Sensitivity.

$$S = \Delta C / a$$

This parameter determines how effectively the device converts mechanical acceleration into an electrical signal. For a perfectly linear sensor, S remains constant across the operating range, and ideally, S=0 when a=0.

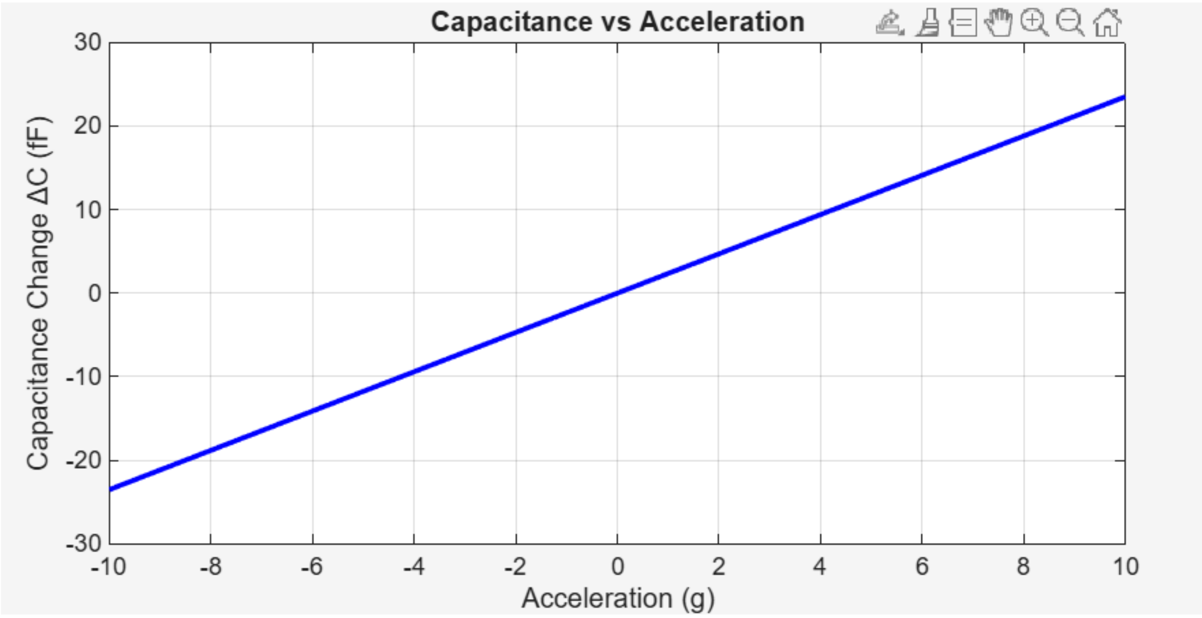

Figure 4: Capacitance vs Acceleration

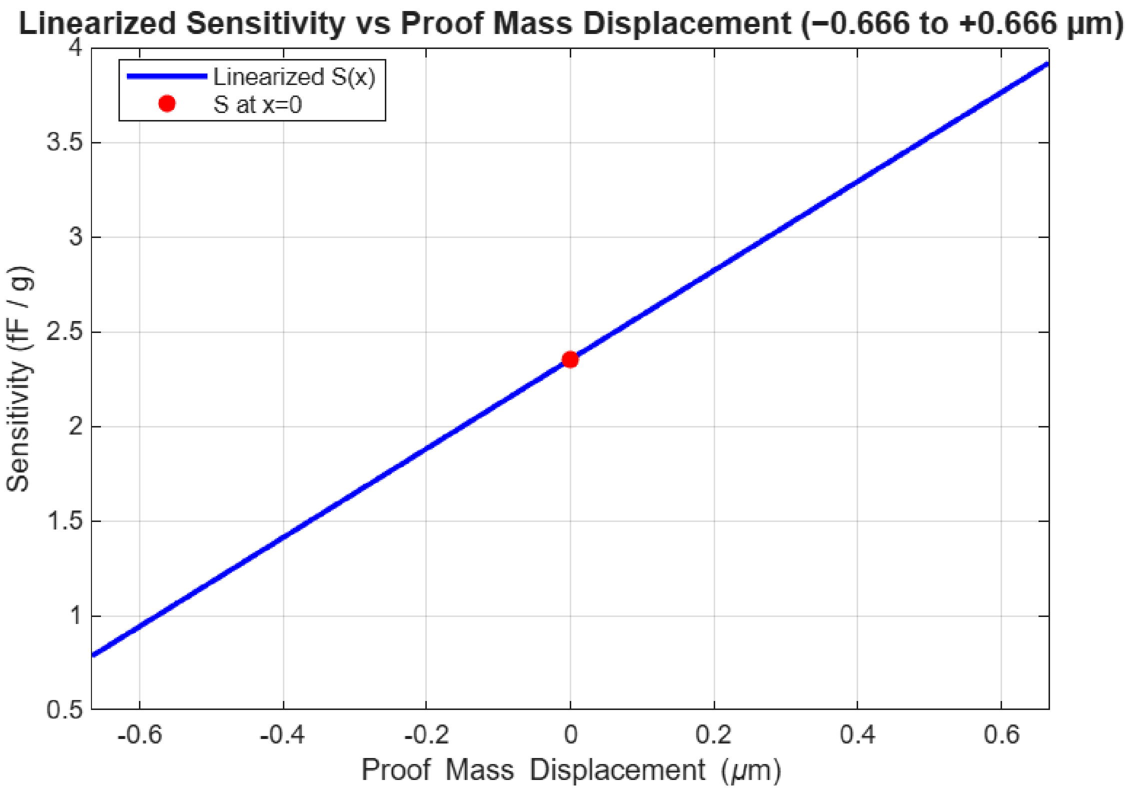

Figure 5: Sensitivity vs Displacement

C. Mechanical Bandwidth

Mechanical bandwidth is the range of frequencies over which a microelectromechanical system (MEMS) accelerometer can accurately measure acceleration without significant signal distortion or loss of sensitivity. It is directly related to the **resonant frequency** of the device and the **damping ratio** of the mechanical structure.

For a comb-type micro accelerometer, the mechanical bandwidth can be approximated as:

$$f_{BW} \approx \frac{f_{res}}{\sqrt{2}}$$

where f_{res} is the resonant frequency of the suspension system.

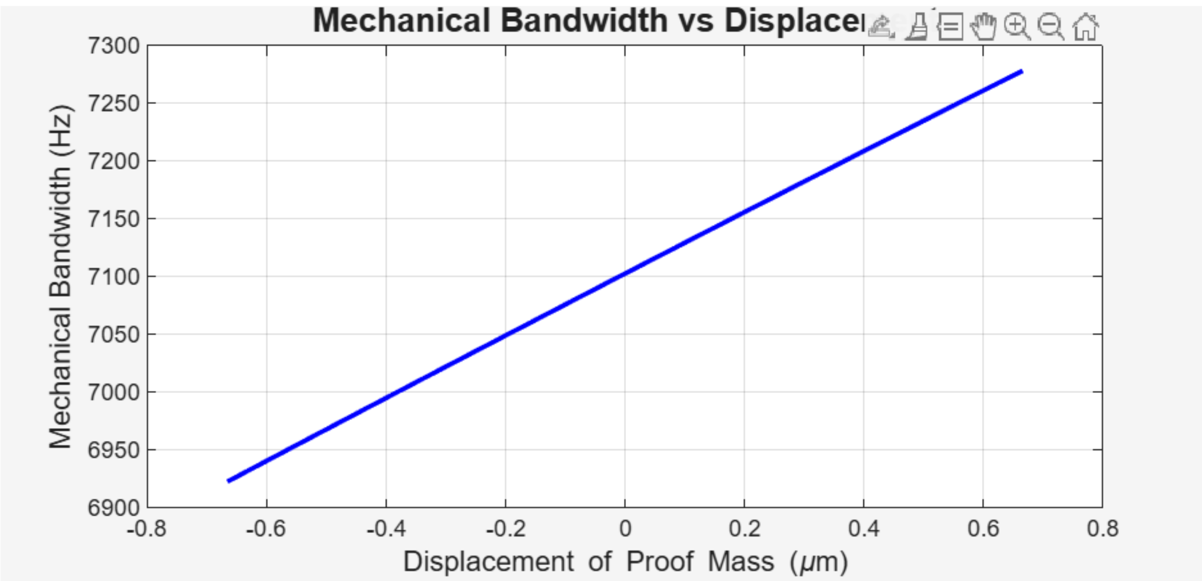

Figure 6: Mechanical Bandwidth vs Displacement

D. Non-Linearity

In a comb-type MEMS accelerometer, *non-linearity* refers to how much the actual capacitance change deviates from the ideal linear relationship between the displacement of the proof mass and the output signal (capacitance or voltage).

Key points for comb-type accelerometers:

- Ideal case: The capacitance change ΔC\Delta CΔC is proportional to the displacement of the proof mass, which is in turn proportional to the applied acceleration.
- Real case: Due to fringe fields, non-uniform gap changes, fabrication imperfections, and mechanical stiffness variations, the relationship deviates slightly from linear.

$$\%\text{Non-linearity} = \frac{\text{Max deviation from ideal line}}{\text{Full-scale output}} \times 100$$

- Effect: At large displacements (near full range), non-linear effects become more pronounced, causing errors in acceleration measurement.

- Measurement: Non-linearity is often expressed as:

- Design goal: Keep non-linearity below a few percent (often < 1%) to ensure accurate sensing.
- Example: If for a ±10 g range, the output capacitance curve bows slightly instead of being a straight line, that curvature represents non-linearity.

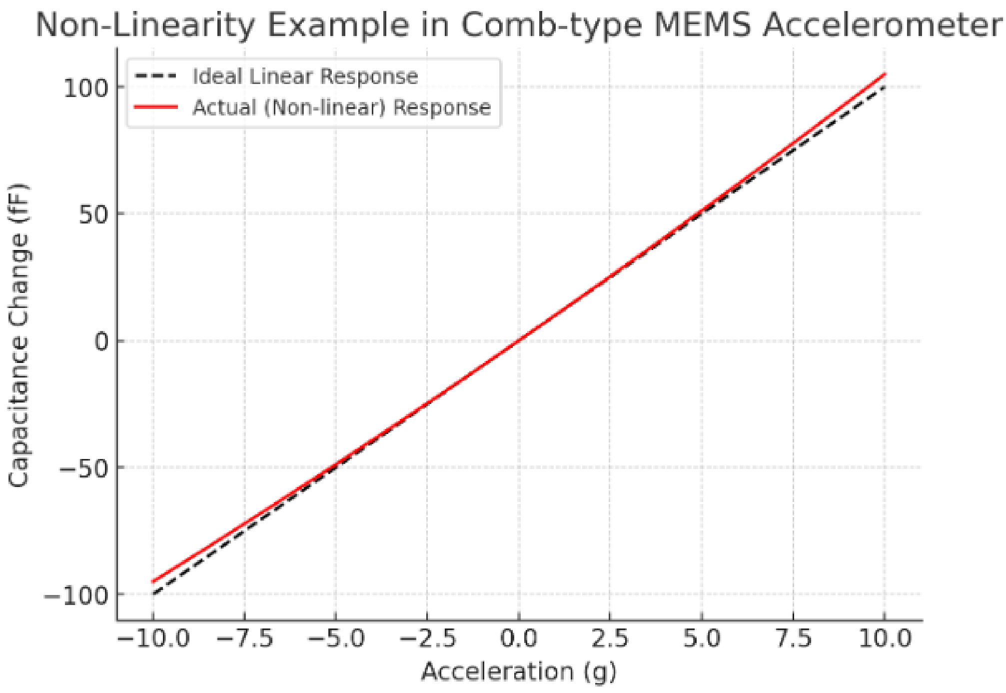

Figure 7: Non linearity for Capacitance vs Acceleration

E. Noise Level

The **noise level** is the smallest acceleration the accelerometer can reliably detect above its inherent noise.

$$a_n = \sqrt{\frac{4k_B T\,\omega_0}{m\,Q}}$$

In comb-type MEMS accelerometers, the **dominant noise source** is often **Brownian motion** of the proof mass, which produces **thermal-mechanical noise**.

The **noise-equivalent acceleration (NEA)** can be expressed as:

Where:

- a_n = Noise equivalent acceleration (m/s²/√Hz)
- k_B = Boltzmann's constant (1.38×10−23 J/K)
- T = Absolute temperature (K)
- ω_0 = Resonant angular frequency ($2\pi f_0$)
- m = Proof mass (kg)
- Q = Mechanical quality factor

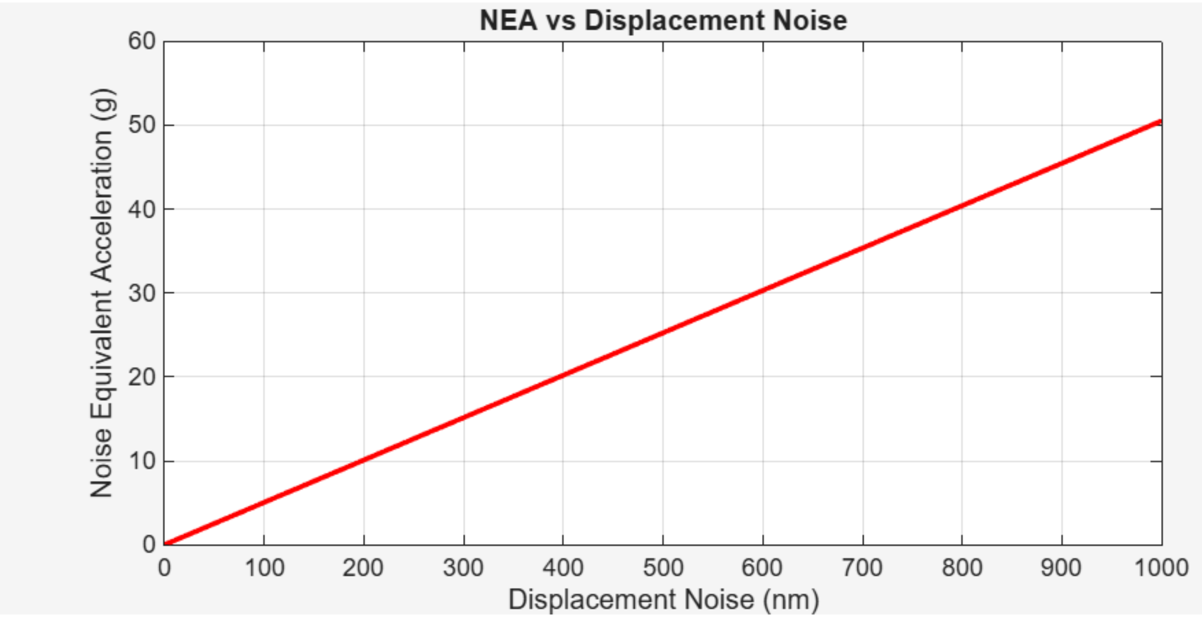

Figure 8: NEA vs Displacement Noise

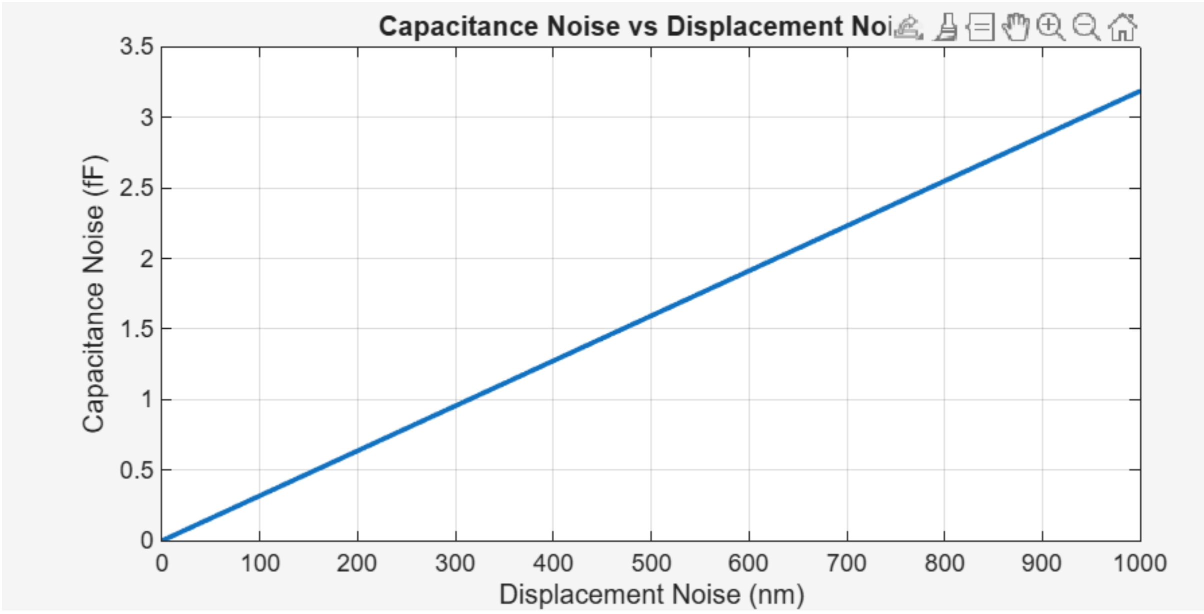

Figure 9: Capacitance Noise vs Displacement Noise

V. Simulation: